

Forecasting August 2026 Food-Shelf Activity in Minnesota

MinneMUDAC 2026 — Question 7: Methods, Validation, and Submission

MinneMUDAC 2026 Undergraduate Team

Information cutoff: July 31, 2026

Abstract

We forecast August 2026 visits (HouseholdsReg), pounds distributed (PoundsReg), and individuals served (IndividualsReg) for each of Minnesota’s 87 counties using only information available on or before July 31, 2026. We first backtested the five reference models specified in the question, then ran a multi-agent study in which four specialist workflows (statistical, panel machine learning, deep learning and foundation models, and structural/data-quality) built 101 model variants that were all scored by one shared, leakage-safe harness. Many sophisticated models beat the seasonal-naive baseline on the historical record, but none did so with statistical significance, and an out-of-sample replay of “pick the best model” performed worse than simple methods. We therefore built the forecast from a small set of robust county-level methods (seasonal naive, damped year-over-year growth, per-county local-model selection, their blend, and a recent three-month mean), letting *each county* use the method that tracked it best over the previous twelve months. Against the plain blend, this county-by-county selection lowers county-level WAPE from 10.7% to 9.5% (visits), 10.8% to 8.9% (pounds), and 10.6% to 9.1% (individuals) across 31 rolling one-month-ahead tests, removes systematic under-forecasting, and is significantly better in paired tests. The submission implies statewide totals of 237,861 visits, 12.99 million pounds, and 738,532 individuals, and it cuts county-level error relative to seasonal naive by about 40%.

1 Task and information cutoff

Question 7 asks for an August 2026 forecast under a strict information cutoff of July 31, 2026, for every internal and external feature, with the prediction pipeline kept in a separate folder and an explicit cutoff check. The submission format requires one row per county with three predicted quantities. Table 1 maps them to the prepared data.

Table 1: Submission targets and their source fields.

Submission column	Prepared field	Source field
HouseholdReg_Predicted	visits	HouseholdsReg
PoundsReg_Predicted	pounds	PoundsReg
IndividualsReg_Predicted	individuals	IndividualsReg

Cutoff enforcement. The cutoff is enforced in code rather than by convention:

1. The loader refuses to run if the source file contains any month after July 2026.
2. Every forecast receives a data slice that ends exactly one month before its target; this is asserted on each of the $\sim 10,000$ model calls in the study.
3. Backtest actuals are looked up only after a prediction has been produced.

- Any tuning (window length, damping, local-model choice) is done inside the slice, using only earlier months; no parameter was tuned against backtest scores.
- No external predictors are used in the submitted model. SNAP, poverty, and Map the Meal Gap variables were excluded because their release dates could not be verified or they are annual and cannot inform a single month.

Data policy. We use the project’s prepared site-month file (`Data/processed/foodshelf_site_month.csv`), which resolves documented duplicates and clear typographical errors. All observations flagged as high outliers or as households exceeding individuals are retained; the policy was locked before model selection and a sensitivity check (Section 2) shows these rows change August growth by less than 0.4 percentage points in 2024–25. Wilkin County has no food-shelf site in the data and is predicted as zero.

2 Data and diagnostics

Statewide monthly totals span January 2022–July 2026 (55 months). Figure 1 shows two regimes: a strong ramp-up through mid-2023 (visits rose from about 124k to 237k per month) and a roughly flat level since 2024.



Figure 1: Statewide monthly outcomes with August values (red), the July 31, 2026 cutoff (dotted), and the August 2026 forecast with an empirical 80% interval.

Is July 2026 complete? Because every model leans on the most recent months, we checked for late or partial reporting. July 2026 had 462 reporting sites (0.99 of the prior three-month mean, inside the normal 0.97–1.04 band), 98.3% of June reporters also reported in July (equal to the 2024–26 average), and the share of sites with sudden drops was at the low end of history. The 8% year-over-year fall in July visits is real: it comes from sites present in both years, and July 2025 was itself an unusual spike driven by about ten sites.

Where growth came from. A shift-share decomposition shows that the 2023 and 2024 growth (+26.4% and +15.3% in visits) was almost entirely same-site growth, not new sites (Figure 2, left). In 2026 year-to-date, visits are up 0.9%, pounds 3.4%, and individuals 4.1%, and the most recent three months show visits down 4.8%.

August seasonality is unstable. The August/July ratio fell from 1.11–1.13 in 2022–23 to 1.03 in 2024 and 0.92 in 2025 (Figure 2, right). The large early ratios were produced by the growth regime, which is why “July times a historical ratio” forecasts perform poorly.

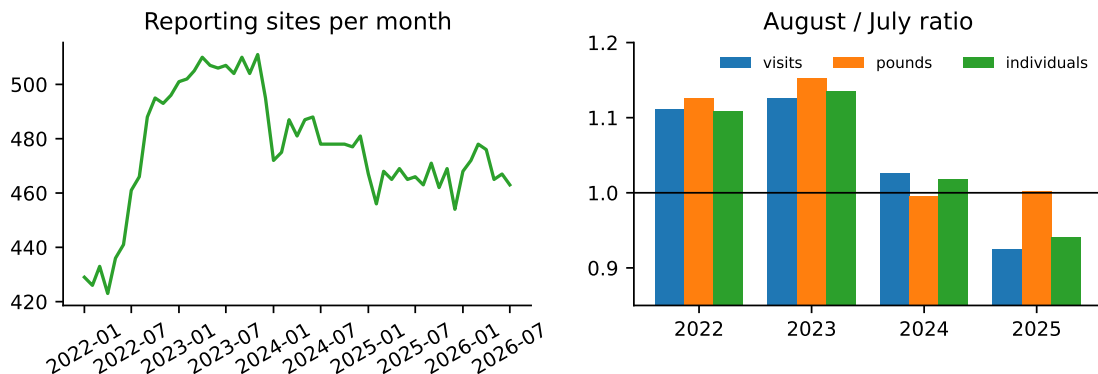


Figure 2: Left: reporting sites per month. Right: August/July ratio by year and outcome.

3 Validation design

Only three historical Augusts can be used as holdouts, too few to choose among many models. We therefore used two protocols with the same one-month horizon as the real task:

Protocol A (official). Forecast August 2023, 2024, and 2025 using data through July 31 of each year.

Protocol B (robustness). Forecast each month from January 2024 to July 2026 one step ahead (31 origins).

The August 2023 fold sits inside the ramp-up regime and has only 19 months of history; it is reported, but we also show the 2024–25 average. Our main error measure is the mean absolute percentage error (MAPE); at county level we use the weighted absolute percentage error $WAPE = \sum_c |\hat{y}_c - y_c| / \sum_c y_c$, because small counties make ordinary MAPE unstable.

Before any agent results were seen, we wrote down a selection rule. A challenger would replace seasonal naive only if it (i) lowered Protocol B MAPE by at least one point, (ii) lowered Protocol A MAPE, (iii) was no more than 1.5 points worse on the 2024–25 Augusts, and (iv) never failed to produce a forecast.

4 The five reference models

Table 2 and Figure 3 report the five models specified in the question, plus the damped year-over-year model (last August $\times (1 + 0.5(g - 1))$, where g is the growth of the latest three months over the same months a year earlier).

Table 2: Reference models: August MAPE (%) over 2023–25, per-fold APE, and Protocol B MAPE.

Outcome	Model	Aug MAPE	2023	2024	2025	Monthly
Visits	Seasonal naive	7.20	19.36	1.75	0.49	8.52
	YTD growth	9.02	0.07	21.33	5.66	9.06
	July $\times$ Aug/Jul ratio	10.14	1.29	8.95	20.19	7.36
	Trend + month regression	13.06	3.77	15.24	20.17	8.72
	Robust median ensemble	9.18	2.53	12.10	12.91	6.01
	Damped YoY	5.28	8.62	3.66	3.57	5.43
Pounds	Seasonal naive	7.03	17.25	0.61	3.23	6.59
	YTD growth	6.68	2.74	16.71	0.60	6.14
	July $\times$ Aug/Jul ratio	9.65	2.32	14.34	12.29	8.71
	Trend + month regression	10.21	4.87	11.19	14.55	8.90
	Robust median ensemble	8.11	3.80	12.77	7.76	6.05
	Damped YoY	5.06	10.06	5.08	0.02	5.16
Individuals	Seasonal naive	10.32	26.15	2.25	2.56	8.75
	YTD growth	9.79	1.79	24.91	2.68	9.79
	July $\times$ Aug/Jul ratio	10.17	2.40	10.23	17.89	8.07
	Trend + month regression	15.49	6.20	17.31	22.97	12.27
	Robust median ensemble	9.45	4.30	13.77	10.28	6.71
	Damped YoY	7.50	15.02	5.34	2.15	5.86

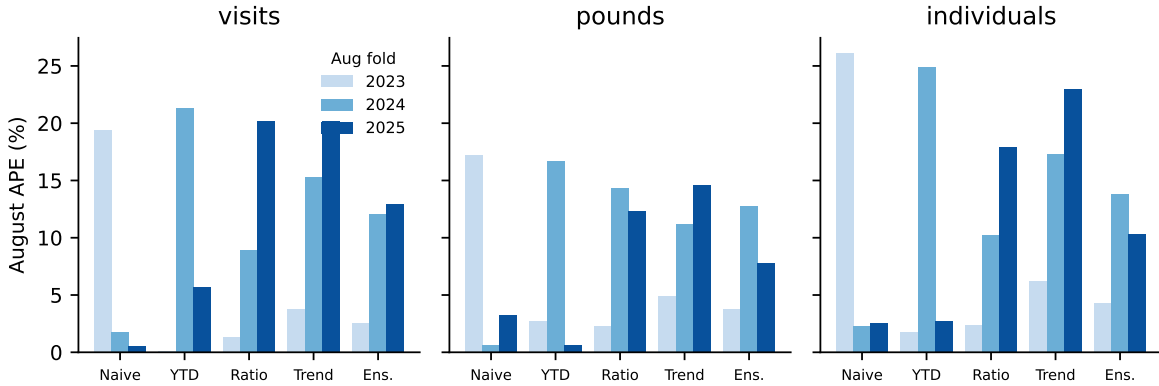


Figure 3: August absolute percentage error of the five reference models in each backtest fold (Naive = seasonal naive, YTD = year-to-date growth, Ratio = July $\times$ Aug/Jul ratio, Trend = trend + month regression, Ens. = robust median ensemble).

The pattern is consistent across outcomes. Growth-based models (YTD growth, ratio, trend) win the 2023 ramp-up fold and then badly over-forecast the flat 2024–25 period, while seasonal naive does the opposite. Damped YoY, which applies only half of recent growth, is the only reference model that avoids large errors in every fold.

5 Multi-agent model study

Four specialist workflows ran in parallel. Each wrote models to the same interface, a function from a cutoff-checked data slice to a statewide (or county) forecast, and was scored by the same harness (Figure 4).

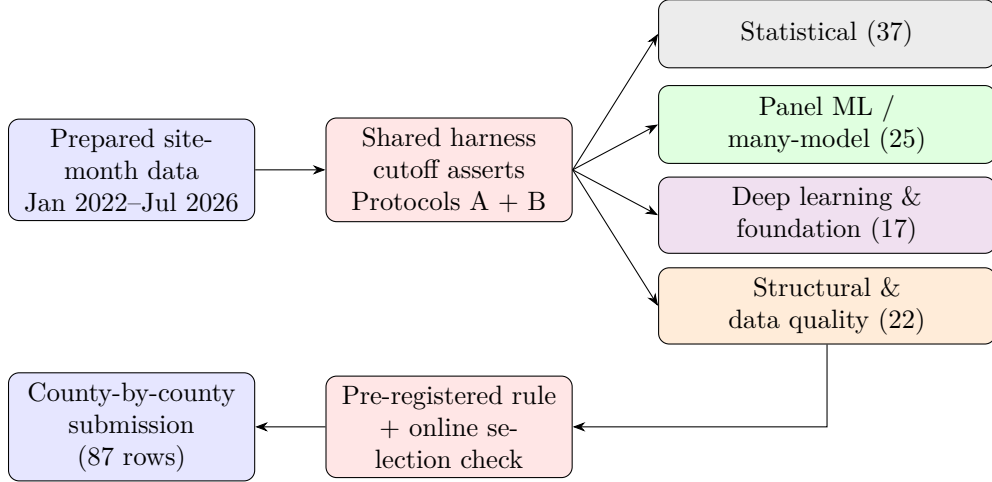


Figure 4: Study workflow. Numbers are the model variants each agent built (101 in total).

Statistical. AutoETS, AutoARIMA, AutoTheta, AutoCES, and MSTL (Nixtla `statsforecast`) on raw and log scales; models of the year-over-year ratio series; a damped-YoY grid; and self-tuning damping chosen by inner rolling evaluation.

Panel ML / many-model. Bottom-up forecasts at the county (87), food-bank (6), and site (~620) levels. These included simple local models, per-unit model selection in the style of the Databricks many-model framework, and global LightGBM and Ridge models trained on scale-free year-over-year log-ratios with past-only features.

Deep learning & foundation. Zero-shot Chronos-Bolt and Chronos-2 on statewide, food-bank, and county series, plus NHITS, NBEATS, and MLP networks trained on the county panel. The networks were retrained on a schedule that never uses data at or after the forecast origin.

Structural & data quality. Same-site growth, models of site entry and exit, reporting sites times per-site averages, cross-metric ratio models (pounds per visit, individuals per visit), and the diagnostics in Section 2.

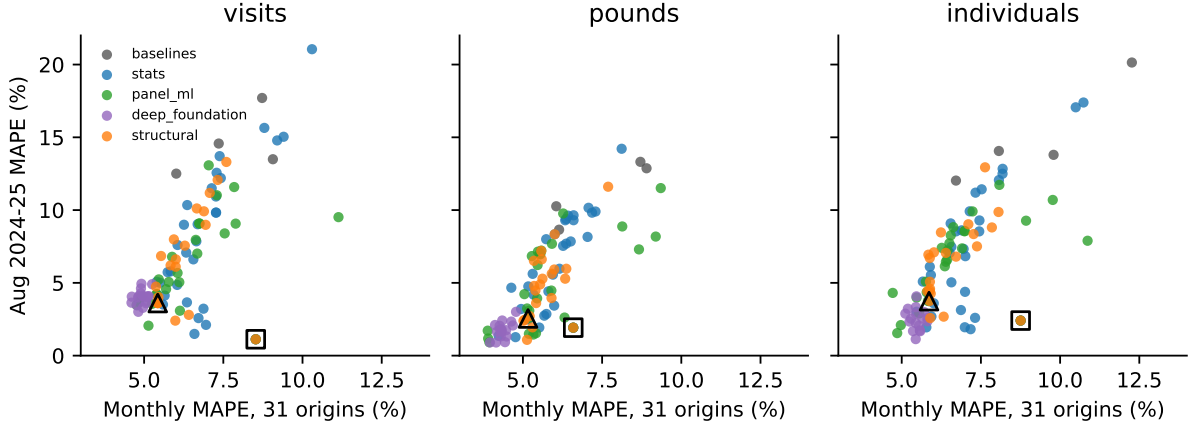


Figure 5: All 101 variants plus baselines. The x-axis is Protocol B MAPE and the y-axis is the August 2024–25 MAPE; lower-left is better. The open square marks seasonal naive and the open triangle damped YoY.

Table 3: Best model from each agent by Protocol B MAPE (%), compared with the baselines.

Agent	Model	Visits	Pounds	Individuals
Baseline	Seasonal naive	8.52	6.59	8.75
Baseline	Damped YoY	5.43	5.16	5.86
Statistical	AutoTheta (log)	4.84	4.23	5.47
Panel ML	Per-county model selection	4.92	3.90	4.86
Deep / foundation	MLP on county panel	4.60	3.96	5.11
Structural	Damped-YoY / cross-metric combo	5.44	5.02	5.81

How much should we trust the winners? The leaders improve on seasonal naive by two to four points, but compared with damped YoY none of the differences is significant (paired tests over 31 origins, all $p > 0.3$). The gains also come almost entirely from 2024, when the series levelled off. To measure selection bias directly, we replayed the selection procedure out of sample: at each month from January 2025 onward, we ranked all candidates using only their earlier errors, averaged the top k , and scored the result (Figure 6). Picking from 101 models was *worse* than using damped YoY or seasonal naive, so most of the apparent improvement is selection luck.

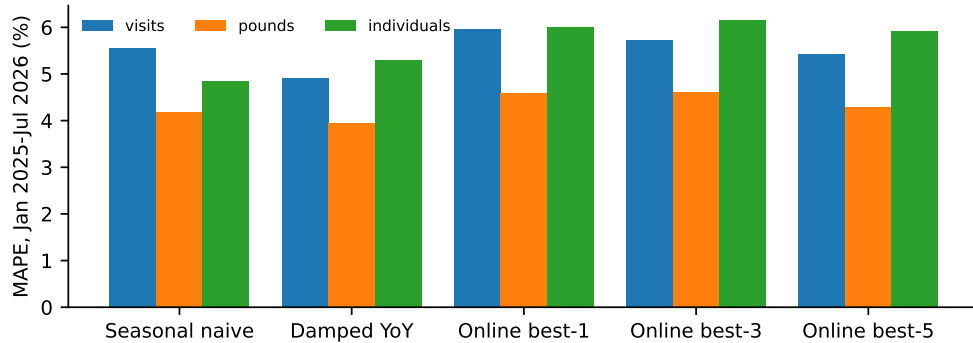


Figure 6: Out-of-sample replay of model selection, January 2025–July 2026 (19 origins).

6 Final model

6.1 From one recipe to county-by-county selection

The study above pointed to simple, robust methods rather than a single sophisticated winner. Our first candidate was an unweighted blend of three such forecasts, applied identically to every county: seasonal naive (last year’s August), damped YoY (last August scaled by half of recent three-month growth), and per-county local-model selection (whichever of five simple local models had the lowest error over the six months before the cutoff).

Because the submission is scored by county, we then examined where the county errors come from. They are concentrated: four counties (Hennepin, Washington, Anoka, Dakota) account for half of all absolute error and fifteen for 80%. The counties forecast worst each had an identifiable site-level event, which one recipe for every county handles poorly:

- **Mille Lacs.** A new site opened in April 2026 with about 1,100 visits per month, roughly tripling the county; seasonal naive still anchors on August 2025 (512 visits).
- **Rice.** A one-off spike in August 2024 (8.1k vs. about 4k normally) became “last August” for the 2025 forecast.
- **Scott and Dakota.** Individual sites with temporary jumps or erratic reporting.

We tested four county-tailoring variants under the same cutoff rules: forecasting each site and summing to counties, the same with spike-capping, and per-county selection with a 12-month window. The best was **county-by-county selection**, which became the final model. For each county it scores five candidates (seasonal naive, damped YoY, local-model selection, the blend, and the recent three-month mean) on that county’s 12 one-month-ahead forecasts before the cutoff, and uses the candidate with the lowest absolute error. All 12 scoring origins lie before the target month. When a county has too little history to score all 12 (the August 2023 fold), the blend is used.

For August 2026 the visits forecasts used the recent mean in 33 counties, damped YoY in 21, the blend in 14, seasonal naive in 10, and local-model selection in 8. Counties with a recent level shift move to the recent mean; for example, Mille Lacs is forecast at 1,550 visits (July 2026: 1,623) instead of the blend’s 991.

6.2 County-level validation

Table 4 compares the final model with the blend on the full set of standard metrics over the 31 monthly origins, and Table 5 breaks county WAPE down by period.

Table 4: County-level metrics over 31 one-month-ahead origins (Jan 2024–Jul 2026): blend → final.

Metric	Visits	Pounds	Individuals
WAPE (%)	10.67 → 9.51	10.75 → 8.91	10.64 → 9.13
County MAPE (%)	15.37 → 13.62	20.28 → 17.36	15.46 → 13.76
Median APE (%)	9.71 → 8.42	10.93 → 9.80	10.11 → 9.10
MAE	306 → 272	15,958 → 13,214	941 → 807
RMSE	1,057 → 970	46,939 → 39,524	3,222 → 2,775
Bias (%)	−3.0 → −0.5	−1.2 → +0.1	−2.7 → +0.1
Statewide APE (%)	5.51 → 4.79	4.72 → 3.71	5.74 → 4.73
Months final better	21 / 31	24 / 31	19 / 31
Wilcoxon p (WAPE)	0.036	< 0.001	0.006

Table 5: County-level WAPE (%) by period. Monthly columns are Protocol B; “2026” covers January–July 2026 only.

Outcome	Model	Aug 2023	Aug 2024	Aug 2025	Monthly 2024–26	Monthly 2026
Visits	Final (submitted)	20.7	8.0	6.1	9.5	12.3
	Blend	20.7	7.4	6.6	10.7	12.3
	Seasonal naive	36.6	10.5	9.4	15.3	13.6
Pounds	Final (submitted)	14.3	7.9	10.5	8.9	10.4
	Blend	14.3	10.2	11.5	10.8	10.4
	Seasonal naive	20.8	11.8	14.9	14.5	11.3
Individuals	Final (submitted)	20.3	8.1	4.1	9.1	11.7
	Blend	20.3	7.5	6.2	10.6	11.7
	Seasonal naive	34.3	10.1	9.3	15.4	13.0

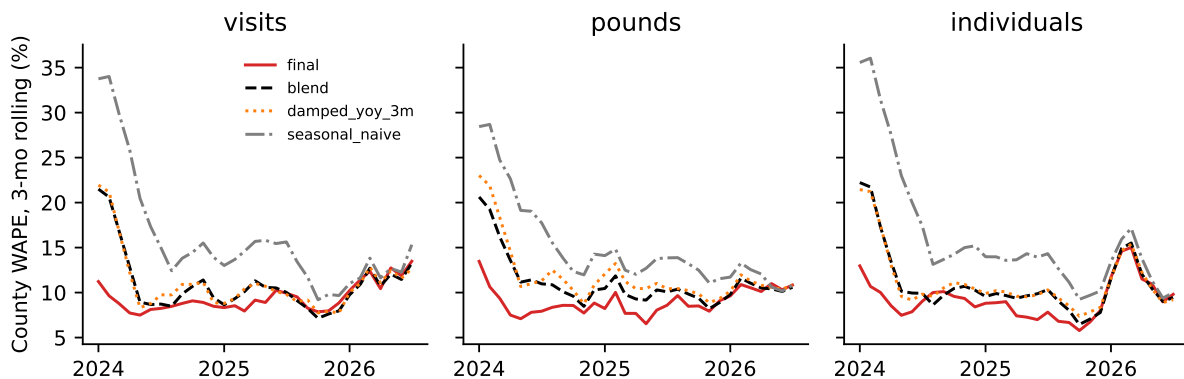


Figure 7: County-level WAPE over the monthly backtest (three-month rolling mean).

The final model improves every metric in Table 4, removes the blend’s systematic under-forecasting, and is significantly better in paired tests. It ties the blend in 2026 and is slightly worse in the August 2024 fold for visits and individuals (+0.6 points). Because it was chosen from four variants after seeing their scores, its advantage is somewhat optimistic; after a correction for the four comparisons, pounds and individuals remain significant and visits becomes borderline ($p \approx 0.14$). Relative to seasonal naive, the final model reduces county error by about 40% (the blend by about 30%).

6.3 Submitted forecast

The 87-row file `Undergraduate_Predictions_Submit.csv` sums to the statewide totals in Table 6. In the August 2023/2024/2025 backtests, the summed county forecasts were within 10.8/1.4/3.8% of the statewide actual for visits, 10.5/1.0/1.9% for pounds, and 15.6/1.2/1.6% for individuals. Figure 8 compares the fifteen largest counties with August 2025.

Table 6: Statewide sums of the submission with empirical 80% ranges (quantiles of the final model’s signed statewide errors over the 31 monthly origins).

Outcome	August 2026 forecast	80% range
Visits	237,861	221,209 – 258,297
Pounds	12,994,439	12,186,624 – 13,615,187
Individuals	738,532	696,924 – 802,298

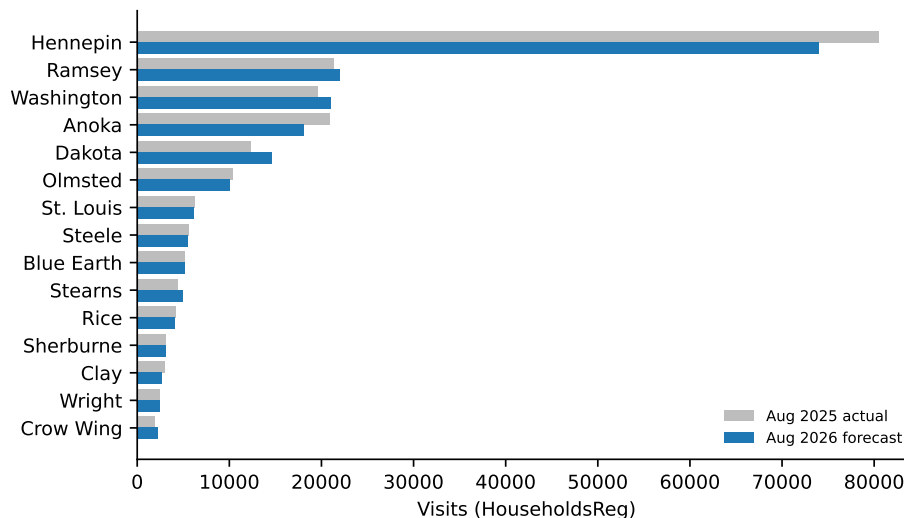


Figure 8: Fifteen largest counties: August 2026 visit forecast vs. August 2025 actual.

7 External predictors tested

We also tested whether external indicators could anticipate county-level changes. Each was evaluated in two ways: applied as a ratio to seasonal naive, and offered as an extra candidate in the county-by-county selection so that a county would use it only where it had worked. Every forecast used only predictor values that would have been published before its target month (Table 7).

Table 7: External predictors (county WAPE over 31 monthly origins unless noted; visits / pounds / individuals).

Predictor	Lag	Result
None (final model)	–	9.51 / 8.91 / 9.13
MN county SNAP enrolment (DHS)	5 mo	With SNAP candidate: 9.49 / 8.81 / 9.11; wins about half the months, $p \approx 0.8$. As a ratio: worse than seasonal naive.
MN DHS SNAP dashboard: persons, inverse persons, inverse benefit per person	2 / 4 mo	No gain: 9.53–9.56 / 8.93–9.00 / 9.11–9.14; pounds significantly worse at lag 2 ($p = 0.014$).
Incoming calls (statewide, effect fitted on past data only)	2 mo	Statewide MAPE 5.6 / 5.5 / 6.2% vs. 5.4 / 3.7 / 5.5% for damped YoY.

None improved the forecast. SNAP enrolment changes by a few percent a year, and the correlation between county SNAP growth and county food-shelf growth is essentially zero, whereas county

food-shelf changes are driven by site-level events. The one large SNAP shock in the sample, the end of pandemic emergency allotments in 2023 (Minnesota benefit per person fell 32%), coincides with the food-shelf surge but occurred only once, so its effect cannot be estimated reliably.

8 Limitations

- There are only three historical August holdouts, and the first lies in a different regime. We add 31 monthly origins as a check, but those are not August-specific.
- With 101 variants tested, small differences are within noise; our recommendation relies on robustness rather than on the single best score.
- The blend and the county-by-county selection were each chosen after looking at results, from small sets of candidates. All candidates and their scores are reported for transparency.
- The data are a single snapshot. Later revisions to July 2026 reports cannot be observed, although every completeness diagnostic was normal.
- The model uses food-shelf history only; external predictors were tested and rejected (Table 7). It therefore cannot anticipate shocks that have not yet appeared in food-shelf data. National SNAP participation fell about 13% year over year by mid-2026 (Minnesota about 4%), yet Minnesota food-shelf visits through July 2026 show no corresponding rise. If reduced SNAP access shifts demand to food shelves with a delay, August 2026 activity could exceed the forecast.

9 Reproducibility

All code is in `forecast_august_2026/`; run the commands below from that folder.

Command	Output
<code>python multiagent/harness.py</code>	reference models (Table 2)
<code>python multiagent/<agent>/run.py</code>	each agent’s predictions and <code>REPORT.md</code>
<code>python multiagent/combine.py</code>	pooled leaderboards and pre-registered selection
<code>python multiagent/honest_selection.py</code>	out-of-sample selection replay
<code>python submission/make_submission.py -team-id ID</code>	final model, county backtest, submission file
<code>python experiments/county_tailoring.py</code>	county-tailoring comparison (Tables 4–5)
<code>python experiments/external_predictors.py</code>	SNAP and call predictor tests (Table 7)
<code>python report/make_figures.py</code>	all figures in this report

Every run writes a `cutoff_check.txt` recording the assertions above.